

Q11

At point P, the displacement, x is negative and velocity, v is negative.

Since $a = -\omega^2 x \Rightarrow F = ma = -m\omega^2 x$, x negative will mean F is positive. B and C are no longer plausible.

Inferring from Fig. 1.2, $a-t$ graph is a cosine graph, hence $v-t$ graph should be a sine graph. Hence A is not possible since its velocity is positive.

Ans: D

Q12